DPUComparator

This DPU tests if its input lies within an interval, outside of the interval or at the interval boundaries.

1. Inputs:

In	Input

2. Dynamic parameters:

Min	Lower interval boundary. Default: -1e300
Max	Upper interval boundary. Default: 1e300
Integer	Should the comparisions be performed in whole numbers? Default: 0

3. Outputs:

Equal	*In* lies within the interval [*Min*, *Max*] (including boundaries)
NotEqual	*In* lies outside of the interval [*Min*, *Max*]
Greater	*In* is greater than *Max*
Less	*In* is smaller than *Min*
GreaterEqual	*In* lies within [*Min*, *Max*] or is greater than *Max*
LessEqual	*In* lies within [*Min*, *Max*] or is smaller than *Min*

4. Functional principle:

The comparision can be performed using the following options:

Normal mode:

- (*Integer* = false)
- In this mode the input *In* is compared with the interval and the outputs are set correspondingly.

Integer mode:

- (*Integer* = true)
- The input *In* is at first converted into a whole number (i.e. the decimal part is ignored). It is then compared with the interval. Thereby, the interval boundaries are also taken as whole numbers.

Setting *Min* = *Max* = z compares *In* with a single number z (and not with an interval).

5. Example:

This example can be used to see if a number (*In*) is positive (*Greater* = 1), negative (*Less* = 1) or zero (*Equal* = 1).

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```
DPUComparator sign
{
	Computer = {LOCALHOST};
	Index = 2;

	Min = 0;
	Max = 0;
};
```